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DRAFT RELATED WORK

Is the Best, the Best? Analysing Best-Paper Awardees' Career Trajectories

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1 Related Work

1.1 Scientific Recognition and the Reward System in Science

A well-known idea in the sociology of science is the *Matthew effect*, the observation that scientists who are already well-known tend to get more credit for their work than lesser-known researchers doing equally good work [1]. The name comes from the Gospel of Matthew: “to him who has, more will be given.” In practice, this means that small early advantages in a research career can compound over time into large differences in visibility, funding, and prestige.

One of the clearest empirical demonstrations of this comes from [2], who looked at a Dutch research funding programme and found that scientists who just barely won a grant went on to attract more than twice as much follow-up funding compared to scientists who just barely missed out, despite being nearly identical in quality at the time of the decision. This kind of threshold effect suggests that formal recognition can have long-lasting consequences that go well beyond the award itself, which is exactly what motivates looking at best paper awards in this thesis.

1.2 Best Paper Awards in Computer Science

Computer science is unusual among academic disciplines in that conferences, not journals, are the main venue for publishing research. Best paper awards at these conferences are therefore a meaningful signal of quality and visibility within the community. [3] has compiled a dataset of best paper awards across 32 major CS conferences going back to 1996, which is the starting point for this thesis.

Some prior work has looked at the citation impact of award-winning CS papers. [4] found that papers selected for awards by programme committees before publication tend to accumulate more citations than those selected after publication. But most of this work focuses on the papers themselves. What is much less studied is what happens to the *people* who wrote those papers, especially the junior authors who may have done most of the work but are just getting started in their careers. That gap is one of the things this thesis tries to address.

1.3 Junior Authors and Early Career Trajectories

Getting the right visibility early in an academic career seems to matter a lot. [5] showed that where you start your career, in terms of institutional prestige, strongly shapes where you end up, which suggests that the academic job market is not purely meritocratic. Along similar lines, [6] found that junior researchers who co-authored with top scientists

early in their careers consistently outperformed peers with similar ability levels who did not have that exposure. Interestingly, this effect was biggest for researchers who were not already at elite institutions, suggesting that external signals of quality can substitute for prestige in some cases.

Does being on a best paper at a major CS conference do something similar for junior authors? It puts their name in front of the whole community, attached to work that has been officially recognised as excellent. Whether that translates into more publications, more citations, or better career opportunities is something that can be tested empirically using bibliometric data, which is what this thesis does.

1.4 Disruption in Science: The CD Index

Beyond counting papers and citations, a more recent strand of research has tried to measure what *kind* of contribution a paper makes, specifically, whether it opens up a new research direction or builds incrementally on existing work. [7] developed the CD index for exactly this purpose. The idea is fairly intuitive, if a paper is genuinely disruptive, papers that cite it will tend to ignore its references (because it made those references less relevant). If a paper is consolidating, citers will engage with both the paper and its references together. Formally:

$$CD = \frac{n_i - n_j}{n_i + n_j + n_k} \quad (1)$$

where n_i counts papers citing only the focal paper, n_j counts papers citing the focal paper and its references, and n_k counts papers citing only the references [7]. Scores range from -1 (fully consolidating) to $+1$ (fully disruptive).

A well-known finding using this measure is that science and technology have been getting less disruptive over time. [8] analysed 45 million papers and 3.9 million patents and found a consistent decline in CD scores across all major fields since the 1940s, including computer science. The explanation is probably a mix of things, fields getting more specialised, papers having longer reference lists, and researchers staying closer to established research agendas. This broader trend is worth keeping in mind when interpreting the CD scores we compute for CS award papers.

1.5 Bibliometric Methods

The main analytical tools used in this thesis are standard in bibliometrics. Difference-in-differences (DiD) analysis compares how a group changes before and after some event, relative to a control group that did not experience the event, it has been used in several

studies of how awards and grants affect research output [2]. Lift analysis is a simpler version of the same idea, just comparing the ratio of post-award to pre-award performance at the author level. For statistical testing, the Wilcoxon signed-rank test is a good fit here because citation and publication counts are skewed and non-parametric tests do not assume normality.

For the CD index specifically, computing it requires pulling citation data from OpenAlex for each paper, finding which papers cited it within five years (CD_5), then checking whether those citers also engaged with the focal paper's references. The five-year window is standard in the literature and works well for CS, where citation cycles tend to be relatively short [8].

1.6 Statement of Purpose

Taken together, the literature points to a clear gap. We know that scientific recognition compounds over time (Matthew effect), that early career visibility matters for junior researchers, and that CS award papers have higher citation impact on average. But we do not know whether being a junior author on a best paper actually accelerates a research career, or whether award-winning papers are doing something distinctive in terms of the kind of contribution they make (disruptive vs. consolidating). This thesis tries to fill both of those gaps using a combination of bibliometric trajectory analysis and CD index measurement across 890 CS best paper awards from 2000 to 2018.

This leads to two research questions:

- RQ1** *Do CS best paper awards accelerate the productivity and citation trajectories of junior coauthors, and does this effect vary by career stage and conference tier?*
- RQ2** *Are CS best paper award papers characterised by higher scientific disruption (CD_5) than comparable non-award papers, and does disruption correlate with the career boost experienced by junior coauthors?*

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